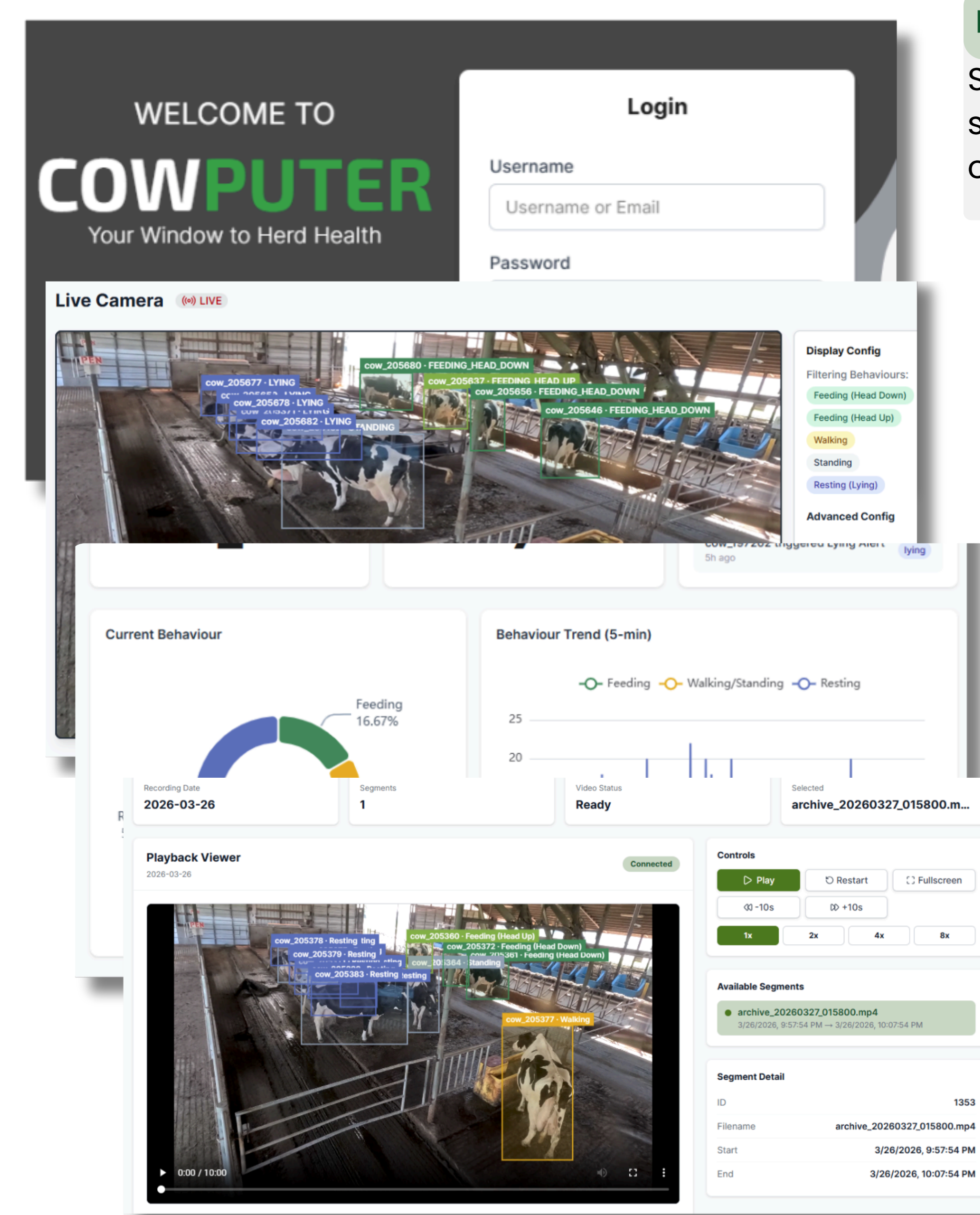
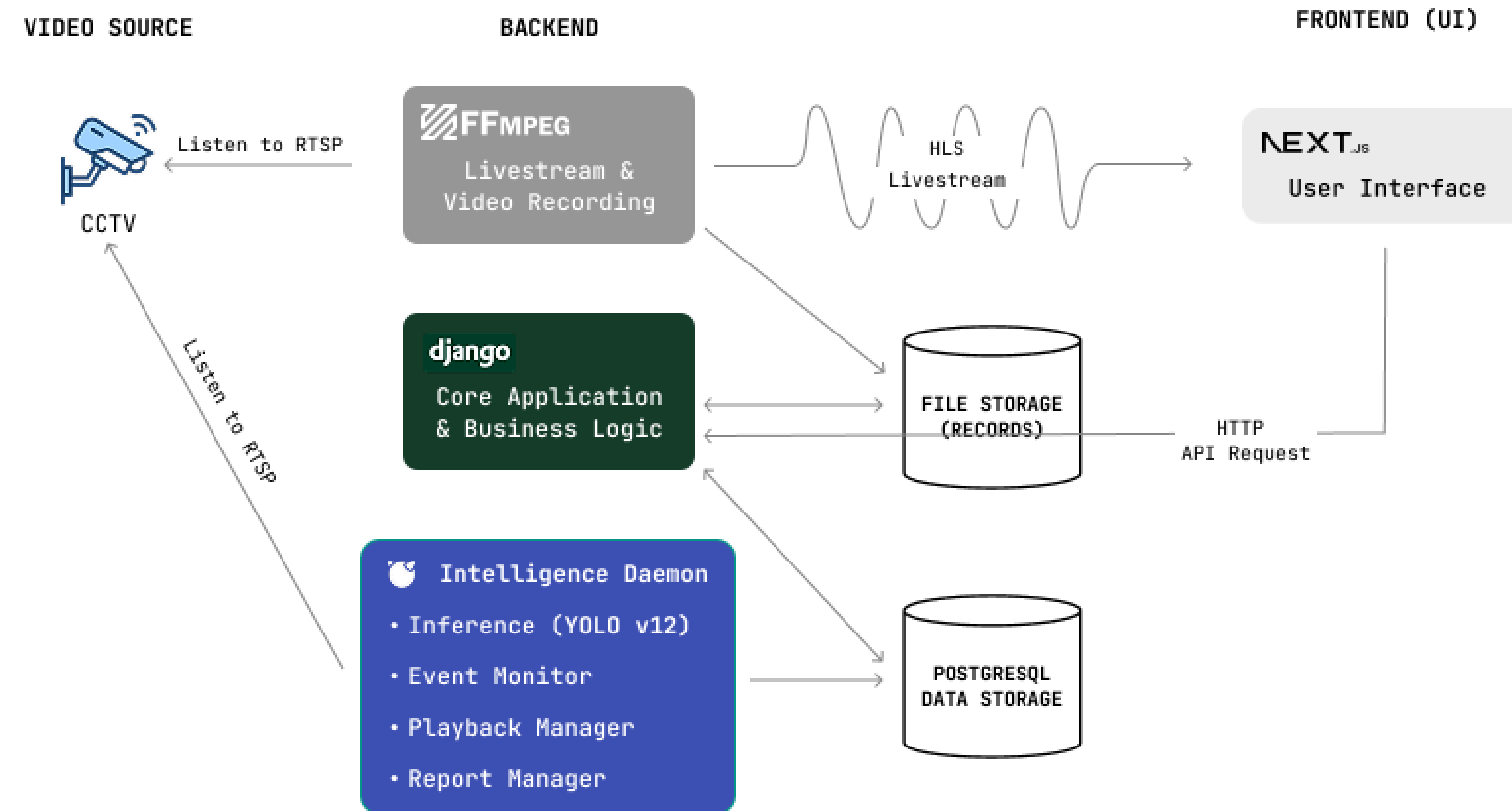


Industry Supervisor: Luke Schuurman, Dr. Amirhossein Sabour



Secure JWT authentication for farm operators, with a first-time setup flow for configuring camera streams and admin credentials.

Real-time barn camera feed with YOLO-powered bounding boxes tracking each cow's behavior. A live stacked area chart and per-cow timeline update every five seconds.

Real-time dashboard showing cow counts, behavior distribution doughnut chart, and a five-minute trend graph. An alerts panel surfaces the latest triggered events with direct links to alert management.

Browse and replay archived recordings by date with color-coded behavior bounding boxes overlaid in sync. Tracks load progressively with interpolated positions for smooth playback.

- **Video Source:** Farm cameras capture live video and stream raw data to the backend via RTSP.
- **FFPMPEG:** Media Server transform the RTSP stream to HLS stream and add machine timestamp for each frame
- **Inference Engine:** The Intelligence Daemon processes incoming frames using a YOLO-based model for cow detection, behavior classification, and tracking. Then store them in the DB.
- **Intelligence Daemon:** Detection results are analyzed for event monitoring, playback management, and report generation.
- **Storage Layer:** Video streams are recorded by the media server and stored in file storage, while structured data (e.g., behaviors, tracking results) is saved in PostgreSQL.
- **Core App:** The Django core application manages business logic and provides APIs to access stored data and processed results.
- **Frontend:** Users interact with the system through the web interface to view live video, playback, alerts, and reports.

- Enhanced detection model — Train a higher-accuracy model with more behavior classes (e.g., mounting, grooming, limping) and improve multi-object tracking to reduce ID switches across frames.
- Health status inference — Correlate prolonged behavioral patterns (e.g., extended lying, reduced feeding) with health indicators to surface early warnings for lameness, heat stress, or illness.
- Multi-camera support — Extend the system to handle multiple barn cameras with unified cross-view tracking and stitched herd-level analytics.
- Mobile notifications — Push real-time alert notifications to farmers' phones via SMS, email, or a companion mobile app.

- [1] H. Vu et al., "MmCows: A Multimodal Dataset for Dairy Cattle Monitoring," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 59451–59467.
- [2] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-Centric Real-Time Object Detectors," *arXiv preprint arXiv:2502.12524*, 2025.